

SET 7

1. R: Create Product, Quantity and Revenue data for 8 products; summarize numeric attributes and export CSV.

The screenshot shows the RStudio interface. The console on the left contains the following R code and its output:

```
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Natural language support but running in an English locale

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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

> # Create product data
> Product <- c("Laptop", "Mobile", "Tablet", "Keyboard",
+             "Mouse", "Monitor", "Printer", "Headset")
>
> Quantity <- c(10, 15, 8, 20, 25, 12, 6, 18)
>
> Revenue <- c(500000, 375000, 240000, 40000,
+             37500, 180000, 90000, 54000)
>
> # Create data frame
> products <- data.frame(Product, Quantity, Revenue)
>
> # Display data
> print(products)
  Product Quantity Revenue
1 Laptop      10  500000
2 Mobile     15  375000
3 Tablet      8  240000
4 Keyboard   20   40000
5 Mouse      25   37500
6 Monitor    12  180000
7 Printer     6   90000
8 Headset    18   54000
>
> # Summarize numeric attributes
> summary(products[, c("Quantity", "Revenue")])
      Quantity      Revenue 
Min.   : 6.00   Min.   : 37500 
1st Qu.: 9.50   1st Qu.: 50500 
Median :13.50  Median :135000 
Mean   :14.25  Mean   :189563 
3rd Qu.:18.50  3rd Qu.:273750 
Max.   :25.00  Max.   :500000
```

The Environment pane on the right shows the 'products' data frame with 8 observations and 3 variables. The 'values' section displays the data for each variable:

Variable	Values
j	3
MAD	6.16326530612245
mean_x	24.1428571428571
normalized	num [1:5] 0.125 0.25 0.375 0.5 0.75
Product	chr [1:8] "Laptop" "Mobile" "Tablet" "Keyboard" "Mouse" "Monitor" "Printer" "Headset"
Quantity	num [1:8] 10 15 8 20 25 12 6 18
Revenue	num [1:8] 500000 375000 240000 40000 37500 180000 90000 54000
x	num [1:7] 14 18 20 23 27 31 36

2. R: Calculate quartiles and IQR for 8,12,14,17,19,22,24,27,30,35,39,44.

The screenshot shows the RStudio interface. The console on the left contains the following R code and its output:

```
R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

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'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

> # Given data
> x <- c(8,12,14,17,19,22,24,27,30,35,39,44)
>
> # Calculate quartiles
> Q1 <- quantile(x, 0.25)
> Q2 <- quantile(x, 0.50)
> Q3 <- quantile(x, 0.75)
>
> # Calculate IQR
> IQR_value <- IQR(x)
>
> # Display results
> cat("Q1 =", Q1, "\n")
Q1 = 16.25
> cat("Q2 (Median) =", Q2, "\n")
Q2 (Median) = 23
> cat("Q3 =", Q3, "\n")
Q3 = 31.25
> cat("IQR =", IQR_value, "\n")
IQR = 15
>
```

The Environment pane on the right shows the 'IQR_value' variable with a value of 15. The 'values' section displays the data for each variable:

Variable	Values
j	3
MAD	6.16326530612245
mean_x	24.1428571428571
normalized	num [1:5] 0.125 0.25 0.375 0.5 0.75
Q1	Named num 16.2
Q2	Named num 23
Q3	Named num 31.2
x	num [1:12] 8 12 14 17 19 22 24 27 30 35 ...

3. WEKA: Create grocery transactions and apply Apriori with minimum support count 3 and confidence 70%; identify strongest rules.

Dataset: Grocery Transactions

```
@relation grocery
@attribute Bread {T,F}
@attribute Milk {T,F}
@attribute Eggs {T,F}
@attribute Butter {T,F}
@attribute Juice {T,F}
@data
T,T,T,F,F
T,T,F,T,F
T,F,T,T,F
T,T,T,T,F
F,T,T,F,T
T,T,F,F,T
F,T,T,T,T
T,F,F,T,T
```

The screenshot shows the Weka Explorer interface with the 'Associate' tab selected. The 'Apriori' algorithm is chosen, and the command line is set to: `Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.375 -S -1.0 -c -1`. The 'Associator output' pane displays the following results:

```
==== Associator model (full training set) ====

Apriori
=====

Minimum support: 0.42 (3 instances)
Minimum metric <confidence>: 0.7
Number of cycles performed: 12

Generated sets of large itemsets:

Size of set of large itemsets L(1): 8
Size of set of large itemsets L(2): 13
Size of set of large itemsets L(3): 3

Best rules found:

1. Juice=F 4 ==> Bread=T 4    <conf:(1)> lift:(1.33) lev:(0.13) [1] conv:(1)
2. Eggs=F 3 ==> Bread=T 3    <conf:(1)> lift:(1.33) lev:(0.09) [0] conv:(0.75)
3. Butter=F 3 ==> Milk=T 3   <conf:(1)> lift:(1.33) lev:(0.09) [0] conv:(0.75)
4. Milk=T Juice=F 3 ==> Bread=T 3    <conf:(1)> lift:(1.33) lev:(0.09) [0] conv:(0.75)
5. Eggs=T Juice=F 3 ==> Bread=T 3    <conf:(1)> lift:(1.33) lev:(0.09) [0] conv:(0.75)
6. Bread=T Eggs=T 3 ==> Juice=F 3   <conf:(1)> lift:(2) lev:(0.19) [1] conv:(1.5)
7. Butter=T Juice=F 3 ==> Bread=T 3   <conf:(1)> lift:(1.33) lev:(0.09) [0] conv:(0.75)
8. Butter=T 5 ==> Bread=T 4    <conf:(0.8)> lift:(1.07) lev:(0.03) [0] conv:(0.63)
9. Eggs=T 5 ==> Milk=T 4     <conf:(0.8)> lift:(1.07) lev:(0.03) [0] conv:(0.63)
10. Juice=T 4 ==> Milk=T 3    <conf:(0.75)> lift:(1) lev:(0) [0] conv:(0.5)
```

4. WEKA: Apply FP-Growth to the grocery data and identify itemsets common to Apriori and FP-Growth.

Dataset: Grocery Transactions – FP-Growth

Use the same Grocery transaction ARFF dataset from Set 7 Q3.

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Associator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.375

Start

Stop

Result list (right-click for ...)

13:39:40 - FPGrowth

Associator output

=== Run information ===

Scheme: weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.375

Relation: grocery

Instances: 8

Attributes: 5

Bread

Milk

Eggs

Butter

Juice

=== Associator model (full training set) ===

No rules found!

Status

OK

Log

x